

Caracterización de las singularidades por la serie de Laurent

Teorema 1 (Tipo de singularidad según la serie de Laurent).

Sea $\sum_{n \in \mathbb{Z}} a_n (z - z_0)^n$ la serie de Laurent de f en la singularidad aislada z_0 , entonces

- (1) z_0 es evitable $\iff \forall n < 0 : a_n = 0$.
- (2) z_0 es un polo de orden $k \in \mathbb{N}$ $\iff \exists k \in \mathbb{N} : a_{-k} \neq 0 \wedge \forall n < -k : a_n = 0$.
- (3) z_0 es esencial $\iff \exists$ infinitos $n \in \mathbb{N} : a_{-n} \neq 0$.

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